

CLARK ENGE

Orange, CA | (714) 906-3643 | clarkenge23@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, Santa Barbara

September 2023 – June 2027

B.S in Data Science and Statistics with Technology Management | GPA: 3.57/4.0

Coursework: Machine Learning, Time-Series, Stochastic Modeling, Experiment Design, Linear Algebra

Universidad Carlos III de Madrid, Spain (Study Abroad)

Fall 2025

Coursework: Statistical Learning, Big Data Analytics, Business Analytics

EXPERIENCE

Sigma Phi Epsilon Gamma Chapter | Vice President of Finance

May 2024 – June 2025

Santa Barbara, CA

- Oversaw \$150,000 annual budget, aligning resource allocation and ensuring accurate financial forecasting.
- Introduced cost optimization measures with external vendors, reducing event-related expenditures by 20%.

Legal Connect | Software Development Intern

June 2024 – September 2024

Chino Hills, CA

- Refactored legacy codebase into modular architecture, improving backend scalability and reducing tech debt.
- Built and deployed modular backend services in C#, improving integration reliability across client-facing workflows

SKILLS

Programming: Python, SQL, R, C#

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, MNE, SciPy, statsmodels

Tools: Excel, Tableau, Power BI, Git, Visual Studio, RStudio, Jira, Microsoft Office

Certifications: IBM Data Science Professional Certificate (Coursera, 2026), SQL Essential Training (NASBA, 2025)

PROJECTS

EEG Sleep Alpha Power Forecasting | SARIMA & Spectral Analysis

May 2026 - Present

Python | MNE | SciPy | NumPy | statsmodels | Matplotlib

- Extracting a ~900-epoch alpha-band power series (8–13 Hz) from PhysioNet Sleep-EDF recordings; fitting SARIMA via ACF/PACF identification and AICc selection with log transformation and seasonal differencing
- Conducting spectral analysis to recover the ~90-minute ultradian sleep cycle; validating forecasts on a held-out test set with back-transformed confidence intervals.

Quantum Error Correction Decoder | ML for Noisy Quantum System

January 2026 – May 2026

Python | Machine Learning | Time-Series Modeling | Simulation | Bayesian Inference

- Built a GRU decoder achieving ~96% accuracy on noisy continuous time-series signals from a 3-qubit system.
- GRU outperformed a Bayesian filter (~94%) and threshold baseline (~86%) under clean conditions and under 4 phases of increasing hardware non-idealities (colored noise, transients, parameter drift);
- Developed hybrid supervision combining periodic ground-truth labels with pseudo-label self-training to sustain accuracy under distribution shift with <2% annotation overhead.

Airbnb LA Real-Estate Revenue Analysis | Big Data Pipeline & Predictive Modeling

December 2025

Python | Pandas | Scikit-learn | EDA | Regression Modeling

- Cleaned and merged 600K+ LA Airbnb listing + calendar rows using Parquet-based ETL, reducing load times by 80%
- Built decision-tree models showing that adding a bedroom increases revenue by 12–17% depending on neighborhood.
- Ranked LA neighborhoods with cross-validated models, highlighting property types that outperformed baselines by 35–50%.